

CD-Dynamax: A JAX-based Python package for continuous-discrete probabilistic state space modeling and inference

Matthew Levine ^{1,2,3¶}, Daniel Waxman ^{2,3}, and Iñigo Urteaga ^{4,5¶}

¹ Broad Institute of MIT and Harvard, USA ² Basis Research Institute, USA ³ Massachusetts Institute of Technology, USA ⁴ BCAM – Basque Center for Applied Mathematics, Spain ⁵ IKERBASQUE — Basque Foundation for Science, Spain ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#))⁰.

Summary

cd-dynamax is a JAX-based Python package for modeling, inference, and learning in *continuous (time dynamics) - discrete (time observation) state space models* (CD-SSMs). It provides a low-level, modular library of CD-SSM models and algorithms for systems where latent states evolve continuously in time according to stochastic differential equations and noisy observations are collected at discrete, possibly irregular, time instants.

cd-dynamax is designed to facilitate adoption, analysis, and experimentation of CD-SSMs by researchers and practitioners in various scientific domains. It provides (i) flexible CD-SSM model definitions of linear and nonlinear dynamics and observation models, (ii) state-of-the-art filtering, smoothing, and forecasting algorithm implementations, and (iii) system identification and model parameter estimation tools, all within a modern, efficient, and autodifferentiable JAX framework.

Mathematical background

Dynamical systems describe complex phenomena in fields including engineering, economics, neuroscience, ecology, and climate science. Continuous-discrete state space models (CD-SSMs) provide a probabilistic framework for systems where noisy observations are collected at irregular intervals while the latent process evolves as a stochastic differential equation (SDE) (Särkkä & Solin, 2019). A CD-SSM is described by:

- A (possibly unknown) stochastic dynamical system, i.e.,

$$dx(t) = f(x(t), u(t), t) dt + L(x(t), u(t), t) dw(t),$$

where:

- $x(t) \in \mathbb{R}^{d_x}$ is the state at time t and $x(0) \sim p(x_0; \varphi_{x_0})$,
- $u(t) \in \mathbb{R}^{d_u}$ is an external input (i.e., control) signal,
- $f: \mathbb{R}^{d_x} \times \mathbb{R}^{d_u} \times \mathbb{R} \rightarrow \mathbb{R}^{d_x}$ is the drift function,
- $L: \mathbb{R}^{d_x} \times \mathbb{R}^{d_u} \times \mathbb{R} \rightarrow \mathbb{R}^{d_x \times d_w}$ is the diffusion coefficient, and
- dw is the derivative of a d_w -dimensional Brownian motion with covariance Q .

- Data observed at arbitrary times $\{t_k\}_{k=1}^K$ via

$$p(y(t_k) | x(t_k), u(t_k), t_k; \varphi_y).$$

Gaussian subclasses recover $y(t_k) = h(x(t_k), u(t_k), t_k) + \eta_k$, where $h: \mathbb{R}^{d_x} \times \mathbb{R}^{d_u} \times \mathbb{R} \rightarrow \mathbb{R}^{d_y}$ and η_k are conditionally independent Gaussian noises.

We denote the collection of all parameters as $\theta = \{f, L, \varphi_{x_0}, Q, \varphi_y\}$.

For discrete-time observations $Y_K = [y(t_1), \dots, y(t_K)]$ and inputs $U_K = [u(t_1), \dots, u(t_K)]$ of a CD-SSM, where inputs are assumed piecewise constant between observation instants, i.e., $u(t) = u(t_k)$ for $t \in [t_k, t_{k+1})$, the main inference and learning objectives are:

- **Filtering:** estimating the distribution of $x(t_K) \mid Y_K, U_K, \theta$.
- **Smoothing:** estimating the distribution of $\{x(t)\}_{t \leq t_K} \mid Y_K, U_K, \theta$.
- **Forecasting:** estimating the distribution of $\{x(t)\}_{t > t_K} \mid Y_K, U_K, \theta$.
- **Parameter learning:** estimating $\theta \mid Y_K, U_K$, either point-wise or in distribution.

Statement of need

cd-dynamax is a JAX-based (Bradbury et al., 2018) open-source Python package that provides a low-level library for inference and learning using CD-SSMs.

Scientists modeling complex dynamical systems must capture continuous-time dynamics while accounting for the discrete and noisy nature of real-world observations. CD-SSMs provide a powerful framework for this, but the associated inference algorithms can be complex, numerically fragile, and computationally demanding, especially for nonlinear CD-SSMs (Murphy, 2023; Särkkä & Solin, 2019; Särkkä & Svensson, 2023).

Existing JAX-native SSM libraries, such as dynamax (S. W. Linderman et al., 2025) and cuthbert (Corenflos & Duffield, 2026), focus on discrete-time state space models and do not address continuous-time dynamics. cd-dynamax fills this gap by providing efficient implementations of state-of-the-art filtering, smoothing, forecasting, and parameter-learning algorithms for CD-SSMs within an autodifferentiable JAX framework that enables gradient-based system identification and modern Bayesian inference.

State of the field

Dynamical systems and state space models (SSMs) are core tools across many scientific disciplines. Python libraries for SSMs (Aicher et al., 2025; Corenflos & Särkkä, 2021; Johnson, 2020; Lee et al., 2023; S. Linderman et al., 2020; Weiss et al., 2024) focus primarily on Hidden Markov Models, discrete-time SSMs, and corresponding Bayesian inference algorithms.

Amongst JAX-native libraries, dynamax (S. W. Linderman et al., 2025), cuthbert (Corenflos & Duffield, 2026), and pfjax (Lysy, 2023) are low-level codebases for inference of discrete-time SSMs. dynestyx (Waxman et al., 2026) is a high-level library that integrates many of these packages, including cd-dynamax, for end-to-end Bayesian dynamical systems inference via NumPyro (Phan et al., 2019).

The rodeo (Wu & Lysy, 2025) library provides JAX-based probabilistic numerics tools for inference under deterministic dynamics (i.e., ordinary differential equations), but does not address stochastic dynamics. The GPJax (Pinder & Dodd, 2022) library provides utilities for linear CD-SSMs, as they may be used to efficiently solve many spatiotemporal Gaussian process problems (Sarkka et al., 2013), but does not include more general CD-SSM methods.

To the best of our knowledge, no existing Python library provides a comprehensive, efficient framework for inference using CD-SSMs.

Software design

cd-dynamax is driven by our vision of modularity, synergies with existing codebases, and extensibility for future research and applications.

We built `cd-dynamax` as a complement and extension to `dynamax` (S. W. Linderman et al., 2025), integrating `diffraction` (Kidger, 2021) for continuous-time dynamics. This lets us build on `dynamax`'s discrete-time SSM tools while supporting continuous-time dynamics, irregular sampling, and nonlinear and non-Gaussian models. By leveraging JAX-native differential equation solvers in `diffraction`, `cd-dynamax` supports hardware acceleration (GPU/TPU) and end-to-end automatic differentiation, enabling efficient gradient-based system identification and modern inference algorithms (e.g., Hamiltonian Monte Carlo).

A key design trade-off was balancing compatibility with the `dynamax` codebase and flexibility in nonlinear CD-SSM model specification and inference. We chose a decoupled architecture where CD-SSM model definitions are separated from inference implementations, such that practitioners can define a linear/nonlinear CD-SSM once and apply varied inference routines. This modular design allows (a) the integration of new model classes and inference algorithms; and (b) extension of the codebase to interact with probabilistic programming interfaces (e.g., NumPyro). `cd-dynamax` is designed to interact with any CD-SSM model (linear or nonlinear) in a unified way for model definition, state-inference, and system-identification, providing a consistently structured, scalable, and extensible framework for CD-SSMs.

We prioritized algorithmic reliability through a test suite that validates correctness against known benchmarks, including CD-SSMs with linear dynamics and discretized nonlinear CD-SSMs. This helps the library serve both methodological researchers extending inference theory and practitioners in systems biology, engineering, and finance who need robust handling of irregularly sampled data.

CD-dynamax modeling and inference

`cd-dynamax` offers:

1. A set of modular CD-SSM model definitions capturing linear and nonlinear dynamics and observation functions — with and without Gaussian assumptions — for irregular, noisy observation times.
2. Low-level, JAX implementations of probabilistic inference algorithms for filtering and smoothing in CD-SSMs (Särkkä & Solin, 2019; Särkkä & Svensson, 2023), including:
 - Kalman filtering and smoothing for linear Gaussian CD-SSMs.
 - Extended Kalman filtering and smoothing for nonlinear CD-SSMs.
 - Unscented and ensemble Kalman filtering for nonlinear CD-SSMs.
 - Differentiable particle filtering for nonlinear CD-SSMs.
3. A high-level interface for constructing and fitting probabilistic SSMs, with functions for:
 - point-estimation of model parameters via gradient-based or black-box optimization (e.g., Scipy (Virtanen et al., 2020), Scipy-jaxopt (Blondel et al., 2021)) of the (approximate) marginal log-likelihood.
 - Bayesian posterior parameter estimation, e.g., Markov Chain Monte Carlo (MCMC) via the BlackJAX (Cabezas et al., 2024) library.

The publicly available `cd-dynamax` documentation and examples describe CD-SSM modeling, filtering, smoothing, forecasting, and fitting for both experts and newcomers.

Research impact statement

`cd-dynamax`'s impact spans two complementary areas: (i) reproducible implementations that achieve state-of-the-art results on CD-SSM inference problems, and (ii) support for active and emerging collaborations in continuous-time dynamical systems modeling and inference.

122 **Benchmarks and reproducible demos**

123 The cd-dynamax repository includes tutorial notebooks and demo scripts demonstrating its
124 ability to handle challenging CD-SSM problems with chaotic dynamics, partial observability,
125 and noisy data:

- 126 ■ **CD-SSM Parameter uncertainty quantification:** fully Bayesian inference of the Lorenz-63
127 system's physical parameters, sampling posterior distributions (via MCMC) from noisy
128 observations.
- 129 ■ **Learning chaotic neural SDEs from noisy observations:** learning unknown nonlinear
130 dynamics from noisy, partially observed data using neural network parameterizations of
131 the SDE drift, demonstrated on the Lorenz-63 system. We showcase accurate short-term
132 forecasts and long-term statistical predictions.
- 133 ■ **Sparse identification of dynamical equations:** leveraging filtering-based likelihoods to
134 perform Bayesian inference of library-coefficients from sparse, noisy trajectory data,
135 demonstrating competitiveness with SINDy and related approaches (Brunton et al.,
136 2016).

137 **Collaborations and community development**

138 cd-dynamax supports ongoing research collaborations and open-source development across
139 multiple groups and use-cases, including:

- 140 ■ Applying CD-SSM inference to patient- and population-level phenotyping via physiologic
141 models with researchers at University of Colorado, Dalhousie University, and Northwestern
142 University. Related studies are in earlier planning stages on populations thought to follow
143 continuous-time dynamics with unknown physiologic parameters.
- 144 ■ Applying CD-SSM inference to study drivers of collective animal behavior from video
145 and acoustic tracking data with researchers at Basis Research Institute.
- 146 ■ Supporting graduate research at the Basque Center for Applied Mathematics (BCAM),
147 including Master's theses on Bayesian modeling and system identification via cd-dynamax.
- 148 ■ cd-dynamax is a crucial back-end support for dynestyx (Waxman et al., 2026), a high-
149 level library for end-to-end Bayesian inference for dynamical systems based on NumPyro
150 (Phan et al., 2019). This validates cd-dynamax's design as a modular, composable
151 low-level library for larger probabilistic modeling ecosystems.
- 152 ■ Serving as an implementation substrate in developing new methods for dynamical systems
153 inference (Waxman, 2025).

154 Several additional collaborations are planned, and we expect them to grow cd-dynamax's user
155 base and collaborative community.

156 **AI usage disclosure**

157 The core architectural design, feature development, and algorithmic implementations are the
158 authors' original work, who carried out all scientific and technical decisions and judgments.

159 We acknowledge the use of generative AI to assist in the development and documentation of
160 cd-dynamax:

- 161 ■ ChatGPT (OpenAI) assisted with code cleanup, plotting scripts, and drafts of docstrings
162 and code comments.
- 163 ■ ChatGPT (OpenAI) and Gemini (Google) assisted with polishing the manuscript's
164 narrative, flow, and clarity.

165 The codebase and manuscript were manually reviewed, tested, and edited by the authors. We
166 verified the correctness of all AI-assisted code, and confirm that the final manuscript accurately
167 reflects our research and design thinking.

Acknowledgements

Iñigo Urteaga acknowledges the support of “la Caixa” foundation’s LCF/BQ/PI22/11910028 award and Grant RYC2023-045922-I by MICIU/AEI/10.13039/501100011033 and by ESF+, as well as funds by MICIU/AEI/10.13039/501100011033 and the BERC 2022-2025 program funded by the Basque Government. Matthew Levine acknowledges support by Basis Research Institute and the Eric and Wendy Schmidt Center at the Broad Institute of MIT and Harvard.

References

- Aicher, C., Putcha, S., Nemeth, C., Fearnhead, P., & Fox, E. (2025). Stochastic gradient MCMC for nonlinear state space models. *Bayesian Analysis*, 20(1). <https://doi.org/10.1214/23-BA1395>
- Blondel, M., Berthet, Q., Cuturi, M., Frostig, R., Hoyer, S., Llinares-López, F., Pedregosa, F., & Vert, J.-P. (2021). Efficient and modular implicit differentiation. *arXiv Preprint arXiv:2105.15183*.
- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018). *JAX: Composable transformations of Python+NumPy programs* (Version 0.3.13). <http://github.com/google/jax>
- Brunton, S. L., Proctor, J. L., & Kutz, J. N. (2016). Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15), 3932–3937. <https://doi.org/10.1073/pnas.1517384113>
- Cabezas, A., Corenflos, A., Lao, J., & Louf, R. (2024). *BlackJAX: Composable Bayesian inference in JAX*. <https://arxiv.org/abs/2402.10797>
- Corenflos, A., & Duffield, S. (2026). *Cuthbert: State-space model inference with JAX*. <https://github.com/state-space-models/cuthbert>
- Corenflos, A., & Särkkä, S. (2021). *Code companion for Bayesian Filtering and Smoothing* (Version 1.0). <https://github.com/EEA-sensors/Bayesian-Filtering-and-Smoothing>
- Johnson, M. J. (2020). *PyHSMM: Bayesian inference in HSMMs and HMMs* (Version 0.0.0). <https://github.com/mattjj/pyhsmm>
- Kidger, P. (2021). *On Neural Differential Equations* [PhD thesis, University of Oxford]. <https://docs.kidger.site/diffraX/>
- Lee, M. A., Yi, B., Martín-Martín, R., Savarese, S., & Bohg, J. (2023). *TorchFilter: Differentiable particle filtering in PyTorch*. <https://github.com/stanford-iprl-lab/torchfilter>
- Linderman, S. W., Chang, P., Harper-Donnelly, G., Kara, A., Li, X., Duran-Martin, G., & Murphy, K. (2025). Dynamax: A python package for probabilistic state space modeling with JAX. *Journal of Open Source Software*, 10(108), 7069. <https://doi.org/10.21105/joss.07069>
- Linderman, S., Antin, B., Zoltowski, D., & Glaser, J. (2020). *SSM: Bayesian Learning and Inference for State Space Models* (Version 0.0.1). <https://github.com/lindermanlab/ssm>
- Lysy, M. (2023). *PFJax: Particle filtering in JAX*. <https://github.com/mlsys/pfjax>
- Murphy, K. P. (2023). *Probabilistic machine learning: Advanced topics*. MIT Press. <http://probml.github.io/book2>
- Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in NumPyro. *arXiv Preprint arXiv:1912.11554*.

- 211 Pinder, T., & Dodd, D. (2022). GPJax: A Gaussian process framework in Jax. *Journal of*
212 *Open Source Software*, 7(75), 4455.
- 213 Sarkka, S., Solin, A., & Hartikainen, J. (2013). Spatiotemporal learning via infinite-dimensional
214 Bayesian filtering and smoothing: A look at Gaussian process regression through Kalman
215 filtering. *IEEE Signal Processing Magazine*, 30(4), 51–61.
- 216 Särkkä, S., & Solin, A. (2019). *Applied stochastic differential equations* (Vol. 10). Cambridge
217 University Press.
- 218 Särkkä, S., & Svensson, L. (2023). *Bayesian filtering and smoothing* (Vol. 17). Cambridge
219 University Press. <https://doi.org/10.1017/CBO9781139344203>
- 220 Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D.,
221 Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M.,
222 Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ...
223 SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing
224 in python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- 225 Waxman, D. (2025). *Sequential Bayesian learning and ensembles for online inference and*
226 *experimental design* [PhD thesis]. State University of New York at Stony Brook.
- 227 Waxman, D., Batenkov, D., Feser, J., Zane, A., Bingham, E., Marzouk, Y., & Levine,
228 M. E. (2026). *Dynestyx: A probabilistic programming library for dynamical systems*.
229 <https://arxiv.org/abs/2606.16985>
- 230 Weiss, R., Du, S., Grobler, J., Cournapeau, D., Pedregosa, F., Varoquaux, G., Mueller, A.,
231 Thirion, B., Nouri, D., Louppe, G., Vanderplas, J., Benediktsson, J., Buitinck, L., Korobov,
232 M., McGibbon, R., Lattarini, S., Niculae, V., Gramfort, A., Lebedev, S., ... Rockhill, A.
233 (2024). *hmmlearn* (Version 0.3.2). <https://github.com/hmmlearn/hmmlearn>
- 234 Wu, M., & Lysy, M. (2025). *Rodeo: Probabilistic methods of parameter inference for ordinary*
235 *differential equations*. <https://arxiv.org/abs/2506.21776>